

Marketing Mix Model

HAMS Data Science Challenge — Solution Report

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1. Problem & Approach

Company X runs an online shop and advertises across 7 paid channels weekly. Marketing actions typically have delayed effects — ads in one week influence sales in following weeks. The goal is to quantify each channel's contribution to revenue and estimate ROI per channel.

A **Bayesian Marketing Mix Model (MMM)** was built using PyMC with NUTS sampling. The Bayesian approach was chosen because it naturally handles uncertainty, produces probability distributions over outcomes rather than single point estimates, and allows incorporation of prior domain knowledge.

Model Formula

revenue = intercept + trend + seasonality + sum(beta[i] x adstock(spend[i], decay[i])) + noise

Component	Description
Intercept	Baseline revenue independent of marketing
Trend	Normalized time index capturing overall revenue direction
Seasonality	Fourier terms (sin/cos pairs) capturing yearly cycles
Adstock	Carry-over transformation applied to each channel's spend
Betas	Channel contribution coefficients (HalfNormal — positive only)
Sigma	Observation noise (HalfNormal)

2. Adstock Carry-Over & Prior Choices

Adstock Transformation

Adstock models the delayed carry-over effect of advertising. Rather than assuming spend only affects revenue in the same week, adstock accumulates spend over time:

$$\text{adstock}[t] = \text{spend}[t] + \text{decay} \times \text{adstock}[t-1]$$

A **decay rate** of 0 means no carry-over (instantaneous effect only). A rate of 1 means infinite memory. Crucially, **decay rates were learned by the model** rather than hardcoded — each channel received its own Beta-distributed decay prior, allowing the data to determine how long each channel's effect persists.

Prior Choices & Justification

Parameter	Prior	Justification
decay[i]	Beta(3, 3)	Symmetric around 0.5 — moderate carry-over expected; constrained 0-1

betas[i]	HalfNormal(1)	Positive only — spend cannot hurt revenue; weakly informative
intercept	Normal(0, 2)	Centered at zero on scaled data; allows flexible baseline
trend_coef	Normal(0, 1)	Allows both upward and downward trends
sin/cos coef	Normal(0, 1)	Weakly informative; lets data determine seasonal strength
sigma	HalfNormal(1)	Positive noise term; weakly informative on scaled data

3. Prior vs Posterior Sampling

Prior sampling reflects our beliefs about parameters before seeing the data. For example, our Beta(3,3) prior on decay rates says we expect moderate carry-over effects, centred around 0.5, with values near 0 or 1 being less likely.

Posterior sampling reflects updated beliefs after the model has processed all 104 weeks of data. Key shifts observed:

Parameter	Prior Mean	Posterior Mean	Interpretation
decay (all channels)	0.50	0.41 – 0.48	Slightly shorter memory than expected
betas[4] (Channel 5)	—	0.249	Strongest channel contributor
betas[0] (Channel 1)	—	0.027	Very weak contributor
trend_coef	0.00	-0.434	Confirms downward revenue trend
sigma	—	0.783	Moderate unexplained variance remains

All R-hat values = 1.00 (one at 1.01) across 4 chains, confirming the posterior distributions are well-converged and trustworthy.

4. Model Performance

Model performance was evaluated using three complementary metrics:

Metric	Value	Interpretation
MAPE (Mean Absolute Percentage Error)	19.63%	Predictions are ~20% off on average — acceptable for MMM
MAE (Mean Absolute Error)	26,577	Average weekly prediction error in euros
R-squared	0.449	Model explains 44.9% of revenue variance
R-hat (Convergence)	1.00	Perfect convergence across all 4 chains

The R-squared of 0.449 is moderate, indicating room for improvement. The primary driver of unexplained variance is the **December 2020 revenue spike** (418k vs mean of 136k) which is difficult to attribute to spend patterns alone and likely reflects an external event or promotion not captured in the data. A MAPE under 20% is generally considered acceptable for marketing mix models.

5. Channel Insights & ROI

ROI per channel was calculated from posterior estimates: attributed revenue (beta x adstocked spend, rescaled) divided by total spend. A ROI above 1.0 indicates the channel generates more revenue than it costs.

Channel	Total Spend	Attributed Revenue	ROI	Verdict
Channel 2	35,739	501,931	14.05	Massively underinvested
Channel 5	891,864	1,364,477	1.53	Profitable
Channel 1	129,543	115,563	0.89	Slightly below break-even
Channel 4	719,174	566,464	0.79	Unprofitable
Channel 6	526,625	310,888	0.59	Unprofitable
Channel 3	2,028,747	1,163,066	0.57	High spend, poor return
Channel 7	2,880,942	917,291	0.32	Worst ROI, highest spend

Key Observations

- **Channel 2 shows the highest ROI at 14.05x** — however this should be interpreted with caution. It has the lowest total spend (35k) of all channels, which can amplify uncertainty in the estimate. The high ROI likely reflects genuine efficiency but warrants further validation with increased spend.
- **Channel 5 is the only other profitable channel (ROI 1.53x)** with substantial spend (891k), making it the most reliable and actionable finding.
- **Channel 7 represents the biggest misallocation** — highest spend (2.88M) but worst ROI (0.32x). Reducing this budget alone could significantly improve overall marketing efficiency.
- **Channel 3 is the second largest spend (2.03M) with poor ROI (0.57x)** — another strong candidate for budget reallocation.

6. Recommendations

Priority	Action	Rationale
1 — High	Scale up Channel 2 spend	Highest ROI — currently massively underinvested at only 35k total
2 — High	Maintain / grow Channel 5	Only channel with confirmed positive ROI at scale
3 — High	Reduce Channel 7 spend	Worst ROI (0.32x) yet consumes 40% of total budget
4 — Medium	Review Channel 3 strategy	Second largest spend with below break-even returns
5 — Low	Investigate Channels 1, 4, 6	All below break-even — optimize or reallocate

What I Would Do With More Time

- Run saturation/diminishing returns modelling (Hill transformation) for high-spend channels
- Add holiday and promotional flags as external regressors to explain the Dec 2020 spike
- Use hierarchical priors to share information across channels
- Cross-validate on held-out weeks to test generalization beyond training data
- Experiment with JAX backend for significantly faster sampling
- Increase Channel 2 spend in a controlled test to validate the 14x ROI estimate